

CS & IT ENGINEERING



'C' Programming

Data Types & I/O Functions

Lecture No.- 01



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Recap of Previous Lecture



Binary operators

- Arithmetic
- Logical (Short circuiting*)
- Bitwise and shift
- Relational
- Assignment



Topics to be Covered



- Ternary operator
- Compound assignment / Short hand assignment
- Operator Precedence & Associativity
- Data Types in 'C'

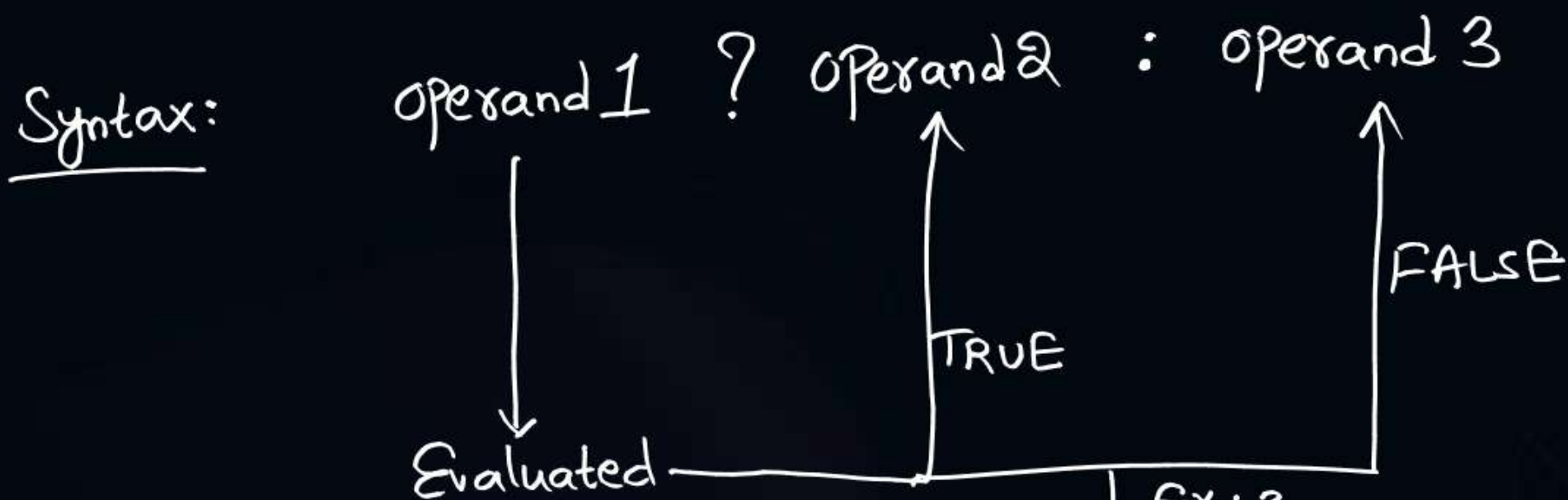




Topic : Ternary Operator



Ternary operator (3 operands) $?:$



- Also known as Conditional operator.

- Associativity $\Rightarrow$ Right to Left

Ex: 1 $a=7, b=3$

```

c = a > b ? a : b ;
    |         ^
    |         | True
    v         |
    c = 7
    
```

Ex: 3

$x=-2, y=5, z=0$

$i = x < y ? y > z ? x ? y ? z ? z : y : x : y : z : x$

True True True False y

$x < y ? y : x$
 $i = y \Rightarrow i = 5$

Ex: 2

$a=1, b=0$

$a ? (b ? \text{Printf}("Hello") : \text{Printf}("Hai")) : \text{Printf}("Bye");$

$b ? \text{Pf}("Hello") : \text{Pf}("Hai")$

$\text{Pf}("Hai")$

$a ? \text{Printf}("Hai") : \text{Printf}("Bye");$

$1 ? \Rightarrow \text{True}$

o/p: Hai

NOTE: Operands cannot be assignment statements, in Ternary operator.

$a = 5, b = 3$

$a > b ? c = 7 : c = 9$

Error



$c = a > b ? 7 : 9$ // $c = 7$

No Error



Topic : Ternary Operator



Short hand Assignment (or) Compound Assignment

Operand operator = Expression

$\Rightarrow$ operand = operand operator (Expression)

Ex: $a * = b \Rightarrow a = a * b$
 $c + = a \Rightarrow c = c + a$
 $x \& = y \Rightarrow x = x \& y$
 $p \cdot = q \Rightarrow p = p \cdot q$
 $i < < = 3 \Rightarrow i = i < < 3$

$a = 7, b = 2, c = 10$
 $c - = a - b$ $\rightarrow c = c - a - b$ ~~Wrong~~
 $\rightarrow c = c - (a - b)$
 $c = \underline{5}$
 $= c - a + b$
 $= 10 - 7 + 2$
 $= 5$



Topic : Operator Precedence & Associativity



- When multiple operators in the same expression, order of Evaluation is decided by Precedence/Priority.
- When Two or more operators with same Precedence, in the expression, order of Evaluation is decided by Associativity.

Ex:

$$w = 17 \gg 3 - \frac{1 * 2 + 1}{2} \& 15 \ll 2 \wedge \frac{14}{4} / 3$$

$$(*, /) > (+, -) > (\gg, \ll) > \& > \wedge > =$$

$$1 * 2 = 2$$

$$14 / 3 = 4$$

$$2 + 1 = 3$$

$$3 - 3 = 0$$

$$17 \gg 0 \Rightarrow 17 / 2^0 = 17 / 1 = 17$$

$$15 \ll 2 = 15 * 2^2 = 60$$

$$17 \& 60 \Rightarrow 17 = 0001\ 0001$$

$$60 = 0011\ 1100$$

$$\underline{\hspace{1cm}} \\ 0001\ 0000 = 16$$

$$0000\ 0100 = 4$$

$$16 \wedge 4 = 20$$

$$\underline{\hspace{1cm}} \\ 16 \wedge 4 \Rightarrow 0001\ 0100 = 20$$

$$w = 20$$



Topic : Operator Precedence & Associativity

BPUAS RBL TAC



Associativity : PTAB (R to L)

Operator Name	Operator	Associativity	Operator Name	Operator	Associativity
<u>Brackets</u>	(), []	Right to Left	<u>Bitwise</u>	& ^	L To R
<u>Postfix</u>	++, --, ., →	Right to Left	<u>Logical</u>	&&	L To R
<u>Unary</u>	++, -- (Prefix), !, ~, + (sign Plus), - (sign minus), size of	Left to Right	<u>Ternary</u>	?:	R To L
<u>Arithmetic</u>	*, /, % +, -	L To R	<u>Assignment</u>	=, +=, -=, *=, /=, /=, &=, ^=, <<=, >>=, --	R To L
<u>Shift</u>	<<, >>	L To R	<u>Comma</u>	,	L To R
<u>Relational</u>	<, <=, >, >=, !=, ==	L To R			

Ex:

$$i = 17 \& 14 \mid 12 \wedge \boxed{\begin{array}{c} 4 * 5 << 3 << \frac{5-3}{2} \\ 160 \end{array}} >> 2 \mid 7 + 4 \cdot 3$$

640
160

$$4 * 5 = 20$$

$$4 \cdot 3 = 1$$

$$5 - 3 = 2$$

$$7 + 1 = 8$$

$$20 << 3 = 20 * 2^3 = 160$$

$$160 << 2 = 160 * 2^2 = 640$$

$$640 >> 2 = 640 / 2^2 = 160$$

$$17 \& 14 = \begin{array}{r} 00010001 \\ 00001110 \\ \hline 00000000 \end{array}$$

$$12 \wedge 160 = \begin{array}{r} 00001100 \\ 10100000 \\ \hline 10101100 \\ 0 = 00000000 \end{array}$$

$$\begin{array}{r} 10101100 \\ 00001000 \\ \hline 10101100 = 172 \end{array}$$

$$i = 172$$



What does the following code do?

```
var a, b: integer;  
begin  
  a:=a+b;  
  b:=a-b;  
  a:=a-b;  
end;
```

Let $a = \overset{3}{\cancel{10}}$, $b = \overset{7}{\cancel{3}}$

$$a = 10$$

$$b = 10 - 3 = 7$$

$$a = 10 - 7 = 3$$

- ✓ A. Exchanges a and b
- B. Doubles a and stores in b ✗
- C. Doubles b and stores in a ✗
- D. Leaves a and b unchanged ✗



The following three 'C' language statements is equivalent to which single statement?

$y = y + 1$; y is incremented before $\Rightarrow$ Pre Increment $\Rightarrow ++y$

$z = \underline{x} + \underline{y}$;

$x = x + 1$ x is incremented after $\Rightarrow$ Post Increment $\Rightarrow x++$

$$\Rightarrow z = (++y) + (x++)$$

a. $z = x + y + 2$;

☒ b. $z = (x++) + (++y)$;

c. $z = (x++) + (y++)$;

d. $z = (x++) + (++y) + 1$;



The attributes of three arithmetic operators in some programming language are given below.

Operator	Precedence	Associativity	Arity (No. of operands)
+	High	Left	Binary
-	Medium	<u>Right</u>	Binary
*	Low	Left	Binary

The value of the expression $2 - 5 + 1 - 7 * 3$ in this language is 9?

$$5 + 1 = 6$$

$$6 - 7 = -1$$

$$2 - (-1) = 2 + 1 = 3$$

$$3 * 3 = 9$$



Which combination of the integer variables x , y and z makes the variable 'a' get the value 4 in the following expression?

$$a = (x > y) ? ((x > z) ? x : z) : ((y > z) ? y : z)$$

~~A.~~ $x = 6, y = 3, z = 5$

~~B.~~ $x = 6, y = 5, z = 3$

~~C.~~ $x = 5, y = 4, z = 5$

~~D.~~ $x = 3, y = 4, z = 2$

$5 > 4 ? \text{True}$ $5 > 5 ? \text{False}$ $a = z \Rightarrow a = 5$

$3 > 4 ? \text{False}$ $4 > 2 ? \text{True}$ $a = y \Rightarrow a = \underline{4}$



1) Data Type

- It defines 3 Parameters of a variable (or) data

1) Nature of data

2) Memory size for data

3) Range of values accepted

To be Contd... 

- Data Types

- Primary (or) Basic (or) Fundamental (or) Primitive : void, char, int, float
- Secondary (or) Derived (or) Non-Primitive : Arrays, Pointers, Structures, Union, File, User-defined (enum, typedef)



2 mins Summary



- Ternary operator
- Short hand assignment
- Operator Precedence & Associativity
- PYQ solving
- Introduction of datatypes.



THANK - YOU